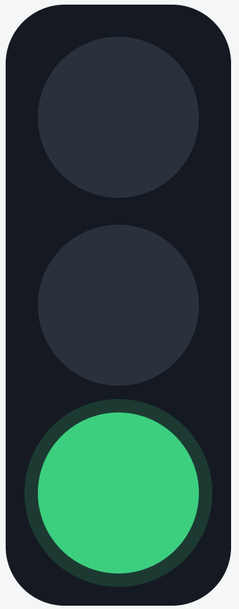
Signal Learner



Stevens Creek backs up all the way to Glenbrook.

An AI that helps a traffic engineer decide when to retime a light.

MONTA VISTA AI CLUB · CUPERTINO

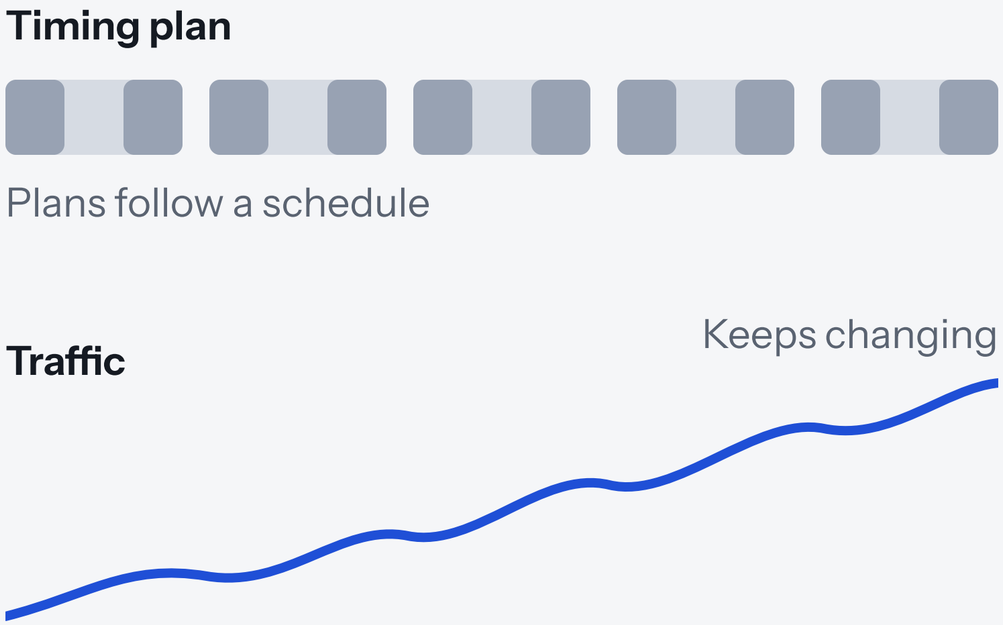
THE PROBLEM

The lights follow scheduled plans. Traffic keeps changing.

> “Stevens Creek Blvd is running time of day plans.”
>
> DAVID STILLMAN, TRANSPORTATION MANAGER, CITY OF CUPERTINO · EMAIL, NOV 1, 2026

WHO IT’S FOR

Cupertino’s traffic engineer, deciding when a road needs new timing.



WHAT WE BUILT

Signal Learner watches a road, practices on a copy, and recommends one change.

1
Watch
Reads the road sensors. Changes nothing.

2
Learn
Learns the traffic and builds a practice copy of the road.

3
Practice
Tries 150 timing plans on the copy.

4
Recommend
One card. The engineer approves or rejects it.

It never touches a real light.

LIVE DEMO

Watch an engineer approve a new timing plan and test it live.

DEMO

Switch to the engineer console.

1

Open the recommendation card

Open the recommendation

2

Approve the change

Approve

3

Test it live on new traffic

Run test on new traffic

Open the engineer console → /console/

RESULT 1 · WOLFE RD, AN UNSEEN TEST ROAD

On a road it had never seen, the delay score dropped 26%.

DELAY SCORE

Extra seconds per trip, compared with an empty road. A trip that never finishes counts as 15 minutes.

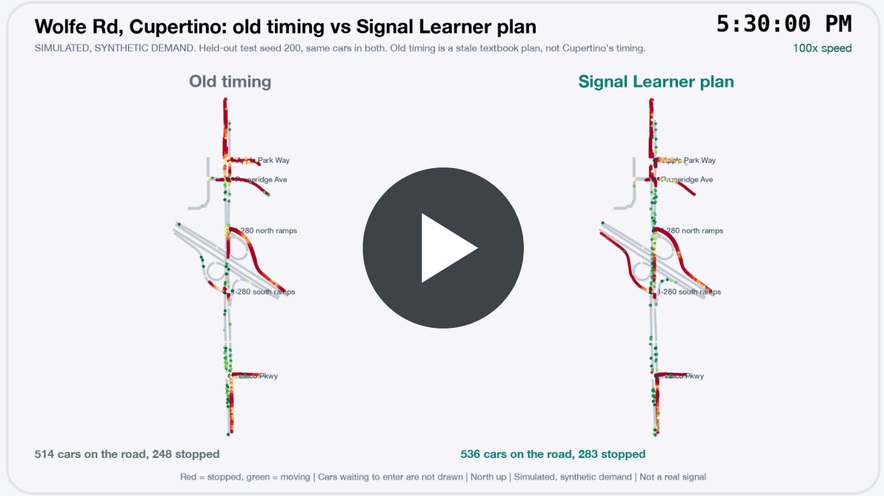
Old timing

870s

Signal Learner

643s

–26%



Watch it: same cars, old timing vs. ours

Better in all 20 test rush hours

RESULT 2 · DE ANZA BLVD, AN UNSEEN TEST ROAD

On De Anza, it said “keep the old timing,” and that was right.

THE CARD SAID

No change

- Keep the old timing

THE PLAN IT TURNED DOWN, RUN ON 20 NEW TEST RUSH HOURS

Old timing

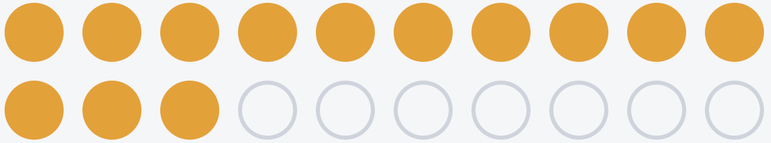


275s

The plan it turned down



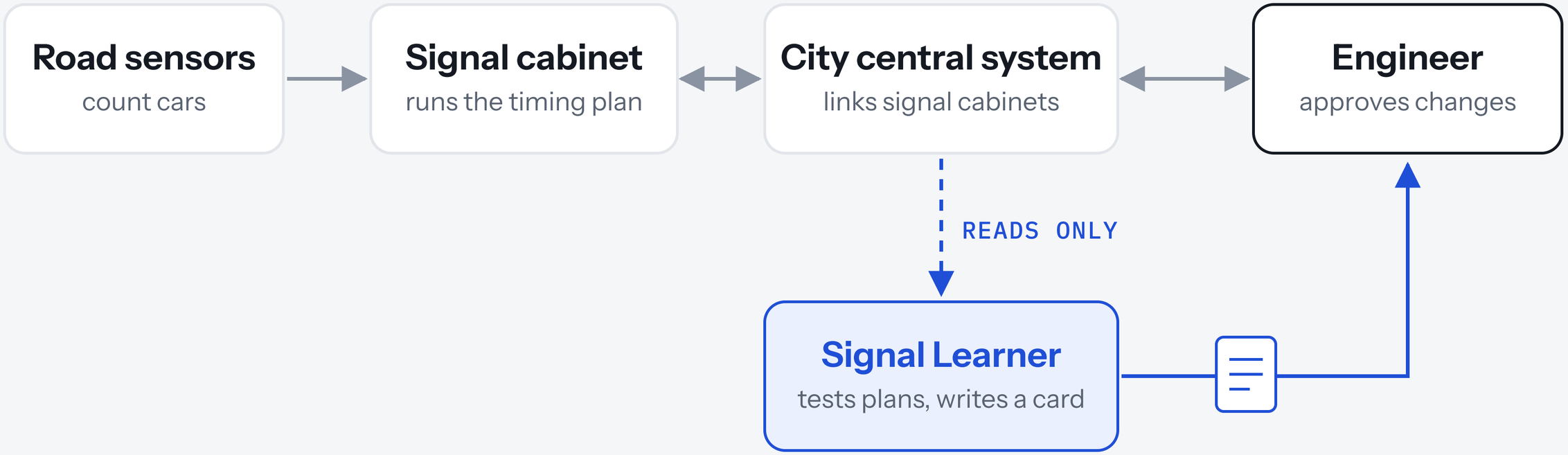
292s 6.4% worse



Worse in 13 of 20

HOW IT FITS THE CITY

Signal Learner only reads. The engineer makes every change.



LIMITS AND NEXT STEPS

Next: real city data and a traffic engineer’s review.

01

Simulation only so far

The test roads used computer-made traffic.

Next: real city counts and timing.

02

Some groups had higher delay scores

On Wolfe, the hardest-hit group’s delay score rose +203 s.

Next: a rule that protects every group.

03

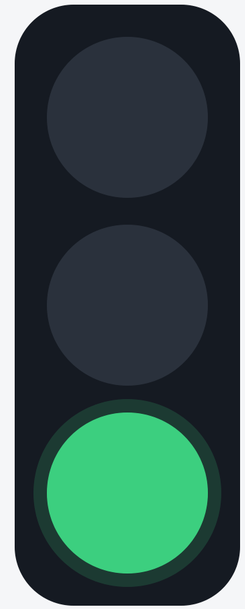
The safety undo was jumpy

On De Anza it undid 4 of 10 harmless changes.

Next: tune it on new roads.

Our ask: one Cupertino traffic engineer.

Review our card with real city timing.



- Wolfe Rd: delay score −26%
- De Anza: kept the old timing, correctly

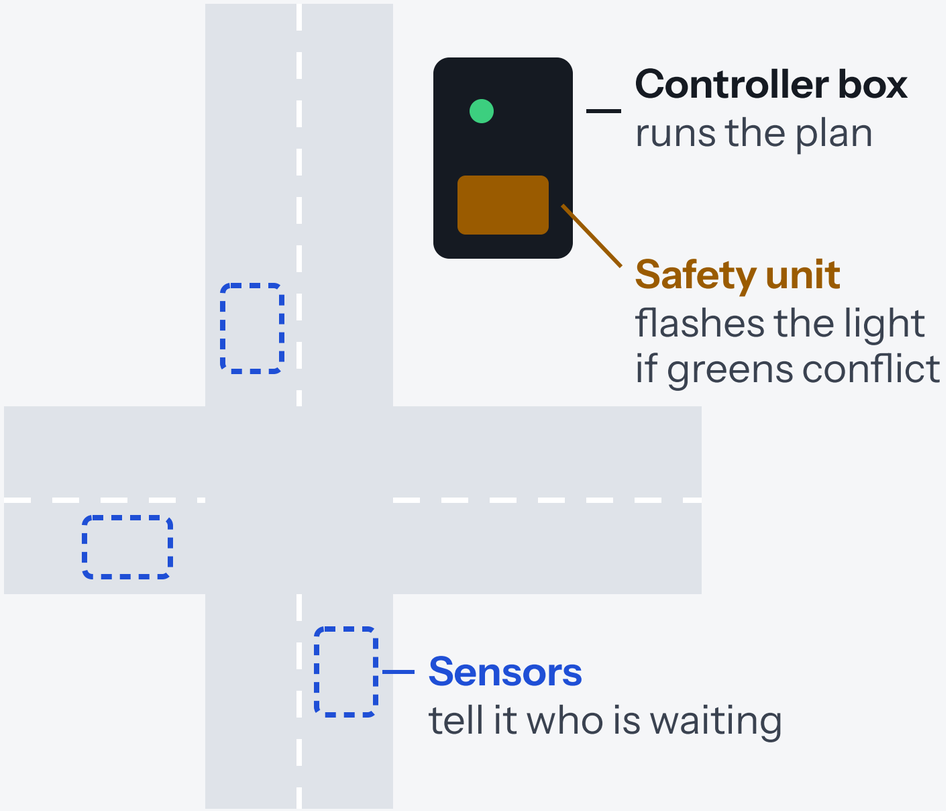
MONTA VISTA AI CLUB

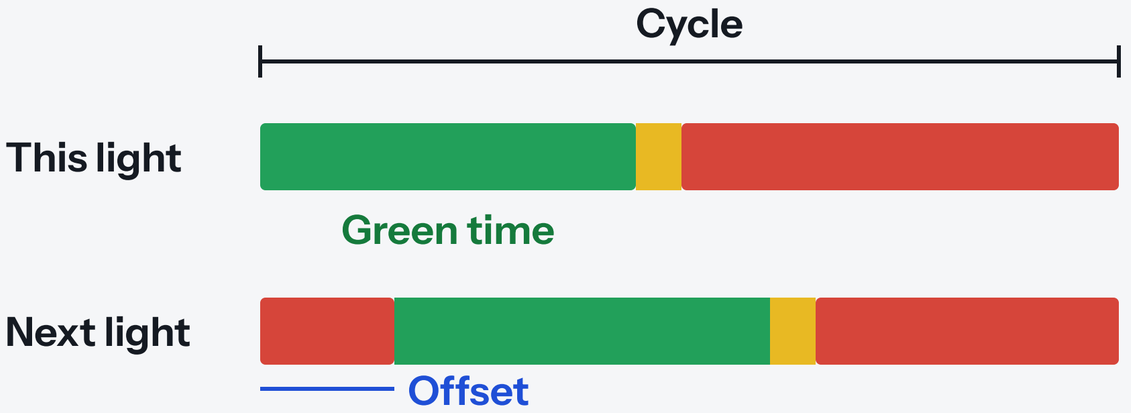
BACKUP

For questions

How a light works · the AI · the fair test · all four roads · existing systems · the delay score · hardware

A small computer at each corner follows a timing plan.





Cycle	One full round of greens.
Green time	How long one direction goes.
Offset	When the next light's green starts, so cars hit green after green.

It learns the traffic pattern, then searches for the best timing plan.

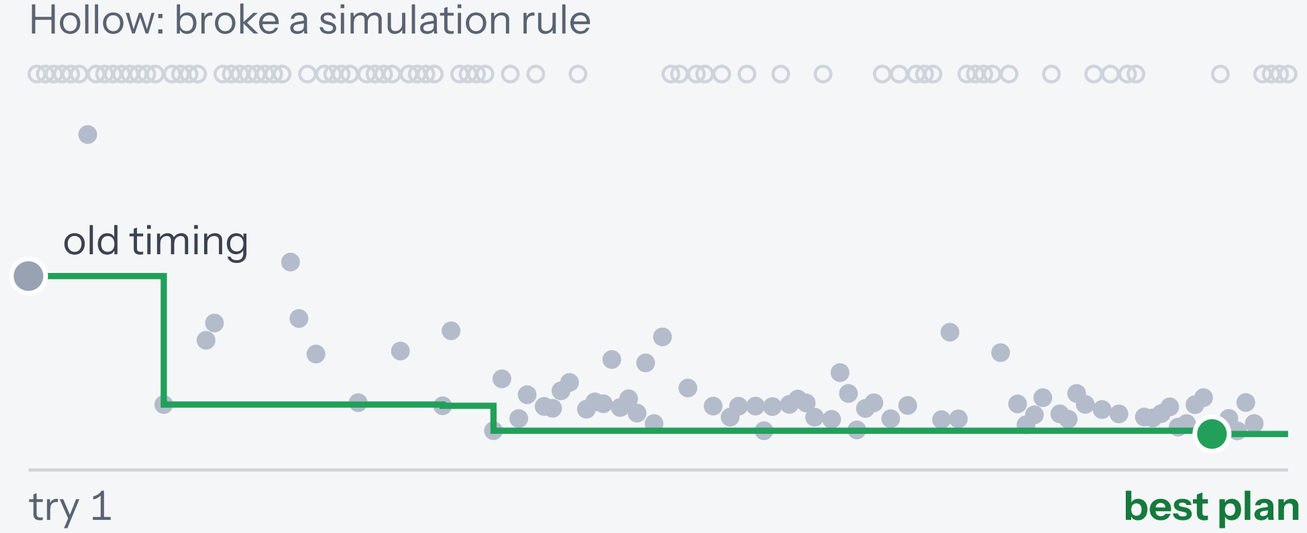
PART 1 · PATTERN LEARNING

Sensor counts become a practice copy of the road

PART 2 · OPTUNA SEARCH

Each new try is picked from how earlier tries scored

150 timing plans tried on Wolfe Rd



EACH DOT IS ONE TRY. LOWER IS BETTER.

It never saw the test trips, and the rules were written first.

WHAT SIGNAL LEARNER GETS

The map, the signals, the old timing and sensor counts



HIDDEN FROM IT

Every test trip and route. A test fails if its code looks.

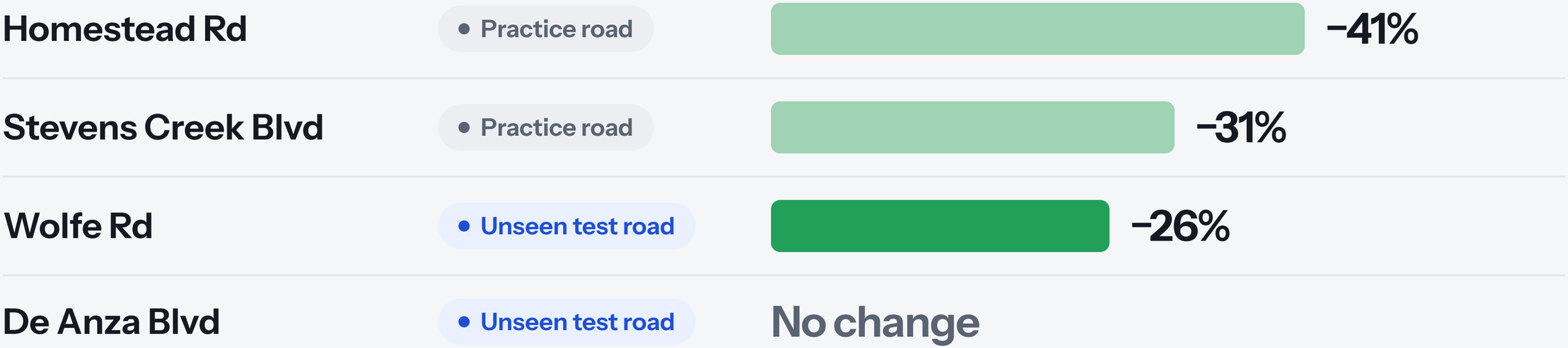
Rules written before the test

Two unseen roads, one try each

Same 20 rush hours for every plan

BACKUP · ALL FOUR ROADS

It lowered the delay score on three roads and correctly left De Anza alone.



Change in delay score. Stevens Creek traffic counts: VTA, 2024 CMP Monitoring and Conformance Report.

BACKUP · EXISTING SYSTEMS

Other systems change lights on the fly or send tips. Ours shows the test behind every card.

CHANGE TIMING ON THE FLY

SCOOT

SCATS

InSync

Surtrac

SynchroGreen

RECOMMEND TO ENGINEERS

Google Green Light

SIGNAL LEARNER

Each card comes with its test

The inputs, the tries and the result.

The delay score is extra time per trip, and stuck trips cost 15 minutes each.

Extra time of every finished trip

VS. AN EMPTY ROAD

+

Each trip that never finished

15 min

+

Each stuck car the simulator removed

15 min

÷

Every trip that was requested

WOLFE RD: TRIPS THAT NEVER FINISHED, PER RUSH HOUR

968 → 217

A cabinet at each corner runs the light, and the city’s central system links them.

Controller

A small computer that steps through the timing plan.

Conflict monitor

A separate safety device. Conflicting greens make the light flash.

Detectors

Wire loops in the pavement, video or radar.

Communications

Links each cabinet to the city’s central system.

Cupertino chose Cubic Trafficware for its central system (2019 announcement).